

BONE STORIES - ACTIVITY 1

Facilitator guide

Find the astragalus

Operational guide for conducting an observational survey in a classroom or museum.

TYPE
Facilitator guide

AUDIENCE
Supervising adults

DURATION
30-60 mins

VERSION
Pilot 1.0

Pilot version - content to be tested and proofread before final release.

1. In brief

The students observe an isolated bone, describe its reliefs, look for its position in the foot then justify their location with at least two clues. The comparison with the sheep or the cow allows us to transfer the landmark without assuming that the members have the same shape.

2. Objectives

- Locate the talus/astragalus in the tarsus region.
- Describe a shape before naming it.
- Link position, joint surfaces and movement.
- Compare humans and other mammals with caution.

3. Expected success

Successful

The tarsus region is identified and two clues are explained.

In progress

The area is plausible, but only one clue is formulated.

To resume

The answer is based on name or chance, without observation.

4. Duration and material

Context	Duration	Material
Class	45-60 mins	Learner sheet, pencils, printed or projected materials.
Museum	30-45 mins	Display case, skeleton, photograph or authorized replica.

5. Quick preparation

À PRÉPARER

Print one sheet per student. Keep the fix out of view. Prepare the vocabulary: foot, tarsus, talus/astragalus, joint, clue.

À DIRE

“Describe first. Then search. Justify with two clues. »

6. Detailed procedure

Time	Adult action	Student action	Purpose
5 mins	Present the bone without giving its location.	Formulate initial ideas.	Bring out representations.
10 mins	Request a description without a name.	Note bumps, hollows, surfaces and symmetry.	Install observation.
10-15 mins	Have the tarsus region searched.	Circle an area and write two clues.	Locate and justify.
10 mins	Distribute animal views.	Compare position and organization of the limb.	Transfer the landmark.
5-10 mins	Discuss the findings as a group.	Write a cautious conclusion.	Distinguish observation and hypothesis.

7. Questions and reminders

To observe

- What do you see without naming the bone?
- Where are the smooth surfaces?
- Which parts seem to fit together?

To justify

- What detail supports your proposition?
- What is missing to be sure?
- Does your second clue confirm the first?

8. Responses or expected elements

- The talus/astragalus is located in the tarsus, between the leg and the rest of the foot.
- Its smooth articular surfaces indicate contacts with other bones.
- In other mammals, a comparable bone exists, but the limb and proportions differ.
- A good answer describes the clues before announcing the location.

9. Differentiation

Need	Adaptation
Help	Limit the search to the area between leg and foot; provide the word bank.
Lighten the writing	Allow an oral or dictated response to a peer.
Go deeper	Compare three mammals and discuss differences in posture.

10. Museum adaptation

Use the sheet as a short station in front of a window, a photograph, a skeleton or a replica. Ask to locate a region of the limb rather than an isolated bone. An image or replica does not entirely replace the original. No heritage object is handled without authorization and protocol from the institution.

11. Quick assessment

Produce an output sentence: "I locate the talus/astragalus in... because I observe... and...". Check the presence of a location and two clues.

12. Common mistakes

À NE PAS AFFIRMER

Don't say that the talus is the visible heel. Do not use a simple silhouette likeness as proof of identification.

- Confuse the talus/astragalus with the calcaneus.
- Show an area without justification.
- Assume that all mammals have a foot identical to ours.

13. Sources and limits

The views of the talus come from BodyParts3D / DBCLS. Animal supports are anatomical images or historical plates credited in the support. The LeConte 1922 plate is old and does not replace a validated modern anatomical diagram. This activity introduces a method of observation; it does not train in expert anatomical identification.

14. Current status and remaining validations

À VÉRIFIER

Sheet and materials ready for a supervised test. Final scientific and educational proofreading, testing with students and physical printing still necessary. A modern anatomical diagram validated by the project remains recommended.

Print plan

Each student: student sheet, all pages.

Adult: guide and correction, one copy.

Projection: observation support.

Color: useful for talus in red; possible grayscale with legends.

Traceability : **Activity page, updates and online credits.**